Worksheet No. - 5

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Aim/Overview of the practical:

To gain hands-on experience in creating and using cursors for row-by-row processing in a database, enabling sequential access and manipulation of query results for complex business logic. **(Company Tags: Infosys, Wipro, TCS, Capgemini)**

Objective:

- **Sequential Data Access:** To understand how to fetch rows one by one from a result set using cursor mechanisms.
- **Row-Level Manipulation:** To perform specific operations or calculations on individual records that require conditional procedural logic.
- **Resource Management:** To learn the lifecycle of a cursor: Declaring, Opening, Fetching, and importantly, Closing and Deallocating to manage system memory.
- **Exception Handling:** To handle cursor-related errors and performance considerations during large-scale data iteration.

Input/Apparatus Used:

- **Oracle**
- **MS SSMS**
- **PostgreSQL**

Code :

```
CREATE TABLE employees (
   emp_id INT,
   emp_name VARCHAR(50),
   salary INT
);
DO $$
```

```
DECLARE
    e_id INT;
    e_salary INT;

    emp_cursor CURSOR FOR
        SELECT emp_id, salary FROM employees;

BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO e_id, e_salary;
        EXIT WHEN NOT FOUND;

        IF e_salary < 50000 THEN
            UPDATE employees
            SET salary = salary + (salary * 0.10)
            WHERE emp_id = e_id;
        END IF;

    END LOOP;

    CLOSE emp_cursor;
END $$;
```

Learning outcomes (What I have learnt):

- Understand how to use **EJB (Stateless Beans)** for implementing business logic in enterprise applications.
- Learn to apply **Bean Validation annotations** to ensure data integrity before storing in the database.
- Gain knowledge of **JPA Entity mapping** and how objects are persisted in relational databases.
- Understand and implement **One-to-One relationship mapping** using @OneToOne and @JoinColumn.
- Develop the ability to write **SQL queries for data analysis**, including joins, aggregation, and handling database-specific errors.